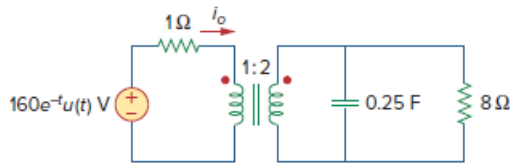


Problem

For the ideal transformer circuit in Fig. 16.94, determine $i_o(t)$.

Figure 16.94:



Step-by-step solution

Step 1 of 8

Refer to the circuit diagram of 16.94 from text book.

The turns ration is, $\frac{N_1}{N_2} = \frac{1}{2}$

The resultant circuit is as shown as below:

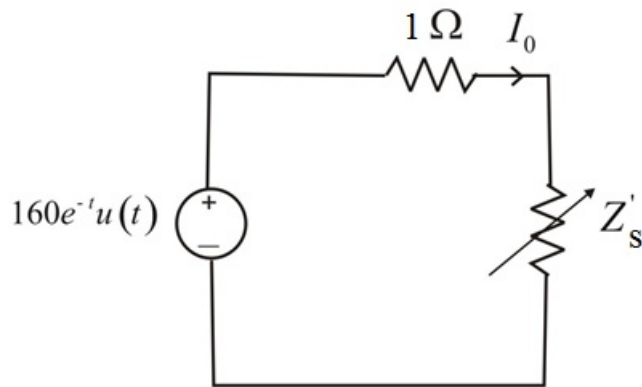


Figure 1

Step 2 of 8

Transfer the impedance on the secondary side to the primary side.

The expression for the transfer of impedance is,

$$Z'_s = K^2 Z_L$$

$$Z'_s = \left(\frac{N_1}{N_2} \right)^2 Z_L \dots\dots (1)$$

Transform the secondary circuit to s-domain and determine the equivalent impedance across the secondary terminals.

The value of the resistor remains the same but the capacitor in s-domain is $\frac{1}{sC}$.

Substitute 0.25 F for C.

$$Z_C = \frac{1}{s(0.25 \text{ F})}$$

Step 3 of 8

The equivalent load impedance is equal to the parallel combination of the resistive impedance and the capacitive impedance Z_C .

$$Z_L = Z_R \parallel Z_C$$

Substitute $8\ \Omega$ for Z_R and $\frac{1}{s(0.25\text{ F})}$ for Z_C .

$$\begin{aligned} Z_L &= (8\ \Omega) \parallel \left(\frac{1}{s(0.25\text{ F})} \right) \\ &= (8\ \Omega) \parallel \left(\frac{4}{s} \right) \\ &= \frac{8 \left(\frac{4}{s} \right)}{8 + \frac{4}{s}} \\ &= \frac{32}{8s + 4} \end{aligned}$$

Step 4 of 8

Substitute $(8\ \Omega) \parallel \left(\frac{1}{s(0.25\text{ F})} \right)$ for Z_L in equation (1).

$$\begin{aligned} Z'_s &= \left(\frac{1}{2} \right)^2 \left[\frac{32}{8s + 4} \right] \\ &= \frac{1}{4} \left[\frac{32}{8s + 4} \right] \\ &= \frac{1}{4} \left[\frac{4}{s + \frac{1}{2}} \right] \\ &= \frac{1}{\left(s + \frac{1}{2} \right)} \end{aligned}$$

Transform the source voltage to s-domain.

$$160e^{-t}u(t) \leftrightarrow 160 \frac{1}{s+1} \quad \left(\because L^{-1} \left(\frac{1}{s+a} \right) = e^{-at}u(t) \right)$$

Replace all the value of the Figure 1 with their s-domain equivalent.

Step 5 of 8

Draw the modified circuit.

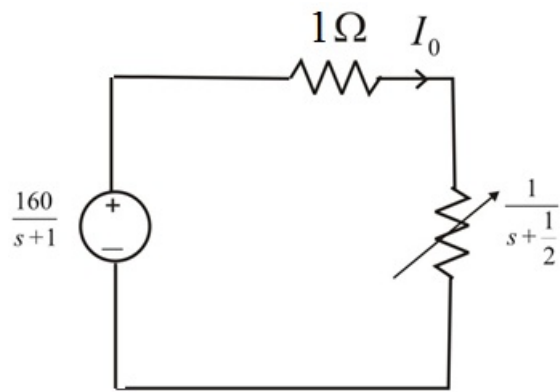


Figure 2

Step 6 of 8

Applying Kirchhoff's voltage law we get

$$\begin{aligned}
 \frac{160}{s+1} &= I_0(s)(1\ \Omega) + I_0(s) \left[\frac{1}{s+\frac{1}{2}} \right] \\
 &= I_0(s) \left[1 + \frac{1}{s+\frac{1}{2}} \right] \\
 &= I_0(s) \left[\frac{s+\frac{1}{2}+1}{s+\frac{1}{2}} \right] \\
 &= I_0(s) \left[\frac{s+\frac{3}{2}}{s+\frac{1}{2}} \right] \\
 I_0(s) &= \left(\frac{160}{s+1} \right) \left[\frac{s+\frac{1}{2}}{s+\frac{3}{2}} \right]
 \end{aligned}$$

Step 7 of 8

Determine the partial fractions of the current $I_o(s)$.

$$I_o(s) = \frac{A}{(s+1)} + \frac{B}{\left(s+\frac{3}{2}\right)} \dots\dots (2)$$

Determine the value of A.

$$\begin{aligned} A &= I_o(s)(s+1) \Big|_{s=-1} \\ &= \left(\frac{160}{s+1} \right) \left[\frac{s+\frac{1}{2}}{s+\frac{3}{2}} \right] (s+1) \Big|_{s=-1} \\ &= 160 \left[\frac{-\frac{1}{2}}{\frac{1}{2}} \right] \\ &= -160 \end{aligned}$$

Determine the value B.

$$\begin{aligned} A &= I_o(s) \left(s + \frac{3}{2} \right) \Big|_{s=-\frac{3}{2}} \\ &= \left(\frac{160}{s+1} \right) \left[\frac{s+\frac{1}{2}}{s+\frac{3}{2}} \right] \left(s + \frac{3}{2} \right) \Big|_{s=-\frac{3}{2}} \\ &= 160 \left[\frac{-1}{-\frac{1}{2}} \right] \\ &= -320 \end{aligned}$$

[Comments \(1\)](#)



Anonymous

it's positive 320

Step 8 of 8

Substitute the value of A and B in equation (2).

$$\begin{aligned} I_o(s) &= \left(\frac{-160}{s+1} \right) + \left[\frac{320}{s+\frac{3}{2}} \right] \\ &= 160 \left[\frac{2}{s+\frac{3}{2}} - \frac{1}{s+1} \right] \end{aligned}$$

Apply inverse Laplace transform we get

$$i_o(t) = 160 \left[2e^{-\frac{3}{2}t} - e^{-t} \right] u(t) \text{ A} \left(\because L^{-1} \left(\frac{1}{s+a} \right) = e^{-at} u(t) \right)$$

Therefore, the circuit current is $160 \left[2e^{-\frac{3}{2}t} - e^{-t} \right] u(t) \text{ A}$.